DOCUMENTATION

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QUESTION NO -1

Introduction

A photography company wants to create a simple image editing tool for their website. The tool should allow users to open, view, edit, and save images with basic operations. This project demonstrates how **OpenCV** and **NumPy** can be used to implement these features.

Objectives

The system should:

- Read and display images
- Save images
- Resize, rotate, flip, and crop images
- Work with image arrays using NumPy
- Split and merge color channels

OOls & Libraries

- **Python** → Programming language
- **OpenCV (cv2)** → Image processing library
- **NumPy** → Numerical operations on image arrays

Implementation with screenshote

[34]:

```
import cv2

# Load image
img = cv2.imread("C:\\Users\\nstip\\Downloads\\doll picture.jpg")

# Display image
cv2.imshow("My Photo", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

[14]:

```
# Save image to disk
cv2.imwrite("saved_photo.jpg", img)
```

[14]: True

[29]:

```
import cv2
import matplotlib.pyplot as plt

resized = cv2.resize(img, (300, 300))

resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)

plt.imshow(resized_rgb)
plt.axis("on")
plt.show()
```

Code

20]:

```python
import numpy as np

# Get image shape
print("Image Shape:", img.shape)

# Access pixel value
pixel = img[50, 50]   # BGR values at row=50, col=50
print("Pixel Value:", pixel)

# Change pixel value
img[50, 50] = [0, 255, 0]   # Green
```

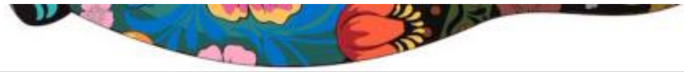
```
Image Shape: (740, 383, 3)
Pixel Value: [  0 255   0]
```

33]:

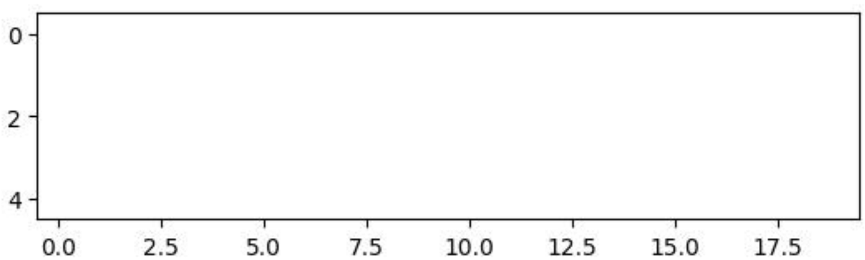
```python
# Rotate
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)
plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.axis("off")
plt.show()

# Flip
flipped = cv2.flip(img, 1)
plt.imshow(cv2.cvtColor(flipped, cv2.COLOR_BGR2RGB))
plt.axis("on")
plt.show()

# Crop
cropped = img[10:15, 20:40]
plt.imshow(cv2.cvtColor(cropped, cv2.COLOR_BGR2RGB))
plt.axis("on")
plt.show()
```




0
100
200
300
400
500
600
700
0
100
200
300


0
2
4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

THANK YOU